THINGS

PEOPLE

Abstract. stuff

1. stuff

Alrighty, let's do a computation!

Definition 1.1. Let I be an ideal in a ring R. Let $f \in R$. The *saturation of I with respect to f* is the set of all ring eleemnts g such that $f^m g \in I$ for some power m. This is usually written as:

$$(I : f^\infty) := \{g \in R : \exists m, f^m g \in I\}$$

Let's look at the following cell, with free variables a, b, c:

$$M(a,b,c) := \begin{bmatrix} a & b & 1 & 0 & 0 & 0 \\ 0 & a & 0 & c & 1 & 0 \\ 0 & 0 & 0 & a & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

This corresponds to the matching $\{(1,6),(2,3),(4,5)\}$. The cut $\{(2,3)\}$ gives the target matching $\{(1,2),(3,6),(4,5)\}$, with target cell matrix

$$P(a,b,c) := \begin{bmatrix} x & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & y & z & 1 & 0 \\ 0 & 0 & 0 & y & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

The pivot flag in the target cell is

$$4, \quad 14, \quad 145, \quad 1456, \quad 12456$$

so we need to use the chart

$$D(P_4^{(1)}) \cap D(P_{14}^{(2)}) \cap D(P_{145}^{(3)}) \cap D(P_{1456}^{(4)}) \cap D(P_{(12456)}^{(5)})$$

$$X = \frac{P_1^{(1)}}{P_4^{(1)}}, \qquad Y = -\frac{P_{124}^{(3)}}{P_{145}^{(3)}}, \qquad Z = \frac{P_{1245}^{(4)}}{P_{1456}^{(4)}}$$

Define $E_{ij} := P_{ij}^{(2)}/P_{14}^{(2)}$ for $\{i,j\} \neq \{1,4\}$. On the target chart, we also have

$$E_{12} = E_{15} = E_{24} = E_{45} = 0$$

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(these can be obtained by taking the nonzero ratios on the source chart, and seeing which ones must die on the target chart)

$$M(a,b,c) := \begin{bmatrix} a & b & 1 & 0 & 0 & 0 \\ 0 & a & 0 & c & 1 & 0 \\ 0 & 0 & 0 & a & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

We calculate the relevant Plücker coordinates:

$$P^{(1)}_1 = a, \quad P^{(1)}_4 = 1$$

$$P^{(2)}_{12} = a^2, \quad P^{(2)}_{14} = -b, \quad P^{(2)}_{15} = a, \quad P^{(2)}_{24} = -a, \quad P^{(2)}_{45} = 1$$

$$P^{(3)}_{124} = -a, \quad P^{(3)}_{145} = 1$$

$$P^{(4)}_{1245} = c, \quad P^{(4)}_{1456} = 1$$

so on the target chart, we have that

$$X = \frac{P^{(1)}_1}{P^{(1)}_4} = a, \qquad Y = -\frac{P^{(3)}_{124}}{P^{(3)}_{145}} = a, \qquad Z = \frac{P^{(4)}_{1245}}{P^{(4)}_{1456}} = c$$

$$E_{12} = -\frac{a^2}{b}, \quad E_{15} = -\frac{a}{b}, \quad E_{24} = \frac{a}{b}, \quad E_{45} = -\frac{1}{b}$$

Define the polynomial ring

$$R := \mathbb{C}[a, b, c, X, Y, Z, E_{12}, E_{15}, E_{24}, E_{45}]$$

on the locus where $b \neq 0$, we have the following equations:

$$X - a = 0, \quad Y - a = 0, \quad Z - c = 0$$

$$bE_{12} + a^2 = 0, \quad bE_{15} + a = 0, \quad bE_{24} - a = 0, \quad bE_{45} + 1 = 0$$

Now consider the ideal consisting of these equations:

$$J := (X - a, Y - a, Z - c, bE_{12} + a^2, bE_{15} + a, bE_{24} - a, bE_{45} + 1)$$

Note: the equation $bE_{45} + 1$ forces $b \neq 0$ on this locus; in general, one should compute $(J : b^\infty)$. Note in this case we have $J = (J : b^\infty)$. Now we compute the elimination ideal:

$$I := (J : b^\infty) \cap \mathbb{C}[X, Y, Z, E_{12}, E_{15}, E_{24}, E_{45}]$$

If we do this by hand: set $q = E_{45}$. Then $q = -1/b$. We have the following equations:

$$E_{15} = Xq, \quad E_{24} = -Xq, \quad E_{12} = X^2 q, \quad Y = X$$

so it turns out, we may replace

$$I = (Y - X, E_{15} - XE_{45}, E_{24} + XE_{45}, E_{12} - X^2 E_{45})$$

in the target cell, we have that $X = Y$. So we have that the intersection

$$\overline{C_{\mathcal{M}}} \cap C_{\mathcal{M}'} = P(a,b,c) := \begin{bmatrix} X & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & X & Z & 1 & 0 \\ 0 & 0 & 0 & X & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

and Z is free.

Proposition 1.2. *Let R be a commutative ring with identity, and $S = R[t]$. Let I be an ideal of R, and $f \in R$. Then, we may compute saturation by the* Rabinowitz trick*:*

$$(I : f^{\infty}) = (I + (1 - tf)) \cap R$$

So in practice, for the above example, we should really compute:

$$\widehat{J} := (X - a, Y - a, Z - c, bE_{12} + a^2, bE_{15} + a, bE_{24} - a, bE_{45} + 1, 1 - tb)$$

and take elimination ideal

$$I = \widehat{J} \cap \mathbb{C}[X, Y, Z, E_{12}, E_{15}, E_{24}, E_{45}]$$

So we just need to figure out how to do this in general!!!!
Questions:

(1) Which E_I's do we need in general?
(2) What do we saturate by in general?

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